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on

Impact of Distraction on the Design of a Driver Monitoring Systems for Advanced Vehicles

Submitted to
Ramaiah Institute of Technology, Bangalore
(Autonomous Institute Affiliated to VTU)
In partial fulfilment of the requirement for the award of degree of

BACHELOR OF ENGINEERING
IN
Electronics and Telecommunication Engineering

For the Academic Year 2024-25

Submitted by

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CERTIFICATE

Certified the project work entitled "**Impact of Distraction on the Design of a Driver Monitoring Systems for Advanced Vehicles**" carried out by Mr. Ayush M (1MS22ET400),Mr. Kishan Kumar N (1MS21ET025),Mr. Shaik Chand (1MS22ET404),Mr. Shri Sudarshan S (1MS21ET049), bonafide student of Ramaiah Institute of Technology, Bangalore, Autonomous institute affiliated to VTU, in partial fulfilment for the award of Bachelor of Engineering in Electronics and Telecommunication Engineering during the year 2024-2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We hereby declare that the project entitled "**Impact of Distraction on the design of a Driver Monitoring Systems for advanced vehicles**" has been carried out independently by us, under the guidance of **Dr. Viswanath Talasila**, Associate Professor, Electronics & Telecommunication Engineering, Ramaiah Institute of Technology, Bangalore. This report has been submitted in partial fulfilment for the award of degree, Bachelor of Engineering in Electronics & Telecommunication Engineering of Ramaiah Institute of Technology (Autonomous Institute, affiliated to VTU, Belgaum) during the year 2024-2025.

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ABSTRACT

The project seeks to develop an innovative system capable of quickly identifying instances of driver distraction while on the road. By leveraging advanced computer vision technology, the system will continuously monitor the driver's actions, meticulously analysing them against a set of predefined behaviours indicative of distraction. These behaviours include, but are not limited to, texting, making phone calls, and looking at a mobile device.

The heart of this system lies in its use of state-of-the-art machine learning algorithms, which enable it to constantly process and interpret the visual data captured by in-car cameras. These cameras will be strategically positioned to provide comprehensive coverage of the driver's actions and surrounding environment, ensuring that even subtle signs of distraction are detected

Machine learning models will be trained on extensive datasets containing various examples of both distracted and attentive driving. This training process will allow the system to accurately differentiate between safe driving practices and potentially hazardous behaviours. The continuous learning capability of these models ensures that the system remains effective over time, adapting to new forms of distraction as they emerge.

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INTRODUCTION

In the modern world, driving has become an essential aspect of daily life, with millions of people relying on vehicles for transportation. However, with the increasing prevalence of mobile devices and other in-car technologies, driver distraction has emerged as a significant threat to road safety. Distraction-related accidents account for a substantial proportion of traffic incidents, often leading to severe injuries and fatalities. Addressing this issue is critical for improving overall traffic safety and reducing the economic and social costs associated with road accidents.

Driver distraction occurs when the driver's attention is diverted away from the primary task of driving due to various factors, such as texting, phone calls, adjusting in-car entertainment systems, or interacting with passengers. These distractions can significantly impair a driver's ability to react to road hazards, maintain proper vehicle control, and make sound decisions while driving. Therefore, there is an urgent need for effective solutions that can detect and mitigate driver distraction in real-time

The advent of advanced technologies, particularly in the fields of computer vision and machine learning, offers promising avenues for addressing this challenge. Real-time detection systems leverage these technologies to continuously monitor drivers' behaviour and identify signs of distraction. By analysing visual data from in car cameras and applying sophisticated algorithms, these systems can recognize specific actions and patterns that indicate distraction, such as looking away from the road, using a mobile device, or engaging in other non - driving activities.

Implementing Impact Of Distraction On The Design Of A Driver Monitoring Systems For Advanced Vehicles involves several key components. These include high-resolution cameras strategically placed within the vehicle to capture comprehensive visual data, robust image processing techniques to extract relevant features, and machine learning models trained on extensive datasets to accurately classify distracted behaviour. Additionally, the system must be capable of processing this information swiftly to provide immediate feedback and alerts to the driver, thereby preventing potential accidents.

1.1 Need for Impact of Distraction on The Design of a Driver Monitoring Systems for Advanced Vehicles:

The need for Impact of Distraction on The Design of a Driver Monitoring Systems for Advanced Vehicles is paramount due to the rising incidence of distraction-related accidents, which are a leading cause of road fatalities and injuries worldwide. The pervasive use of mobile devices and in-car entertainment systems has exacerbated the problem, significantly impairing drivers' ability to focus and react to road hazards. Addressing this issue is critical for enhancing road safety, as distracted driving leads to severe consequences, including collisions and pedestrian accidents.

Advances in computer vision, machine learning, and in-car camera technologies now enable the development of systems capable of real-time monitoring and analysis of driver behaviour, providing a feasible solution to detect and mitigate distractions. Such systems are essential not only for improving individual driver safety but also for aiding regulatory compliance, as many regions have laws against using handheld devices while driving. Furthermore, reducing distraction-related accidents can significantly lower the economic and social costs associated with traffic incidents, including medical expenses, lost productivity, and emotional trauma. Integrating Impact of Distraction on The Design of a Driver Monitoring Systems for Advanced Vehicles with existing driver assistance systems can create a comprehensive safety ecosystem, enhancing the overall performance and safety of modern vehicles. This technological advancement aligns with consumer demand for safer driving experiences and positions automakers competitively in the market. Additionally, as the automotive industry transitions toward fully autonomous vehicles, ensuring that human drivers remain attentive during manual driving phases is crucial. for Impact of Distraction on The Design of a Driver Monitoring Systems for Advanced Vehicles thus represent a proactive approach to road safety, leveraging cutting edge technology to address one of the most pressing challenges in modern transportation.

1.2 Motivation

The motivation to build a system for Impact Of Distraction On The Design Of A Driver Monitoring Systems For Advanced Vehicles is multifaceted and compelling. Primarily, the aim is to enhance road safety by reducing the number of accidents caused by distracted driving, which is a major contributor to traffic incidents and fatalities. Advances in computer vision, machine learning, and sensor technologies now make it feasible to develop sophisticated real-time monitoring systems that can detect and alert drivers to potential distractions. This technological capability, coupled with stringent legal and regulatory measures against distracted driving, underscores the need for such a system. Moreover, the economic impact of traffic accidents, including medical expenses, vehicle repairs, and lost productivity, highlights the potential cost savings of preventing distraction-related incidents. Consumer demand for advanced safety features further motivates the development of this technology, as it can enhance vehicle appeal and market competitiveness. Additionally, as the automotive industry transitions towards semi-autonomous and fully autonomous vehicles, ensuring driver attentiveness during manual driving phases remains crucial, making real-time distraction detection an essential interim solution. The ethical and social responsibility to reduce human suffering associated with traffic accidents also drives this initiative, promoting a safer driving environment for all road users. Furthermore, implementing such a system can generate valuable data on driving behaviors, informing further research and policy-making to continuously improve road safety. Ultimately, developing Impact Of Distraction On The Design Of Driver Monitoring Systems For Advanced Vehicles positions developers and automakers as leaders in automotive safety innovation, contributing to setting new industry standards and fostering a safer, more efficient transportation ecosystem.

1.3 Application

The implementation of for Impact Of Distraction On The Design Of A Driver Monitoring Systems For Advanced Vehicles has numerous applications that extend across various domains within the automotive and transportation sectors:

1. **Passenger Vehicles:**
   - In consumer cars, Impact Of Distraction On The Design Of Driver Monitoring Systems For Advanced Vehicles can provide immediate feedback to drivers. By monitoring the driver’s behaviour and alerting them when they become distracted, these systems can help prevent accidents, enhancing the overall safety of everyday driving.

2. **Commercial Fleets:**
    - For fleet management companies, such systems can be integrated into trucks, delivery vans, and other commercial vehicles. Fleet managers can monitor driver behaviour in real-time, ensuring that their drivers adhere to safety standards. This not only helps in reducing accident rates but also in lowering insurance costs and improving operational efficiency.

3. **Ride-sharing Services:**
    - Ride-sharing companies can implement distraction detection systems to ensure the safety of their passengers. By ensuring that drivers remain focused on the road, these systems can help ride-sharing services maintain high safety standards and build trust with their customers.

4. **Public Transportation:**
    - Buses, trains, and other forms of public transport can benefit from real-time distraction detection systems to ensure that operators remain attentive while driving or conducting. This can prevent accidents in densely populated urban areas and enhance the overall safety of public transportation.

5. **Autonomous Vehicles:**
    - In semi-autonomous and autonomous vehicles, Impact Of Distraction On The Design Of Driver Monitoring Systems For Advanced Vehicles can serve as a crucial backup by ensuring that human drivers are ready to take control when needed. This is particularly important in Level 2 and Level 3 autonomous systems, where the vehicle can handle some driving tasks but still requires human oversight.

6. **Insurance Industry:**
    - Insurance companies can use data from distraction detection systems to assess driver behavior and adjust premiums accordingly. Safe driving discounts and other incentives can be offered to drivers who demonstrate consistent attentiveness, promoting safer driving habits.

7. **Law Enforcement and Regulatory Bodies:**
    - Impact Of Distraction On The Design Of Driver Monitoring Systems For Advanced Vehicles can assist law enforcement agencies in monitoring and enforcing distracted driving laws. Data collected from these systems can be used to support regulatory compliance and aid in accident investigations by providing concrete evidence of driver behaviour.

8. **Driver Education and Training:**
    - These systems can be used in driving schools and training programs to educate new drivers about the dangers of distraction and to reinforce attentive driving habits. Real-time feedback during training sessions can help new drivers develop safer driving behaviours from the start.

9. **Vehicle Manufacturers:**
    - Automakers can integrate distraction detection technology into their vehicles as a standard safety feature. This can enhance the appeal of their vehicles, meet regulatory requirements, and differentiate their brand in the competitive automotive market.

10. **Road Safety Research and Development:**
    - Researchers and developers can use data collected from distraction detection systems to study driving behaviours and develop new safety technologies. This data-driven approach can lead to continuous improvements in road safety measures and the development of innovative solutions.

1.4 **Limitations**:

Despite its potential benefits, Impact Of Distraction On The Design Of Driver Monitoring Systems For Advanced Vehicles faces several limitations that need to be addressed. One major challenge is the accuracy and reliability of the detection systems. Advanced computer vision and machine learning algorithms require extensive training on diverse datasets to accurately identify distracted behaviours, yet they may still struggle with variations in lighting conditions, driver appearances, and in-car environments. Additionally, false positives and false negatives can undermine the system's effectiveness, leading to either unnecessary alerts that could irritate drivers or missed detections that fail to prevent accidents. Another limitation is privacy concerns. Continuous monitoring of drivers raises significant issues regarding data privacy and consent. Ensuring that the system adheres to strict privacy standards and gains driver consent is crucial to prevent misuse of the captured data.

Moreover, the implementation cost of these systems can be prohibitive, particularly for budget-sensitive consumer markets and smaller fleet operators. The high costs of advanced cameras, sensors, and processing units may limit widespread adoption.

Technical integration with existing vehicle systems also poses a challenge. Retrofitting older vehicles with new technology can be complex and expensive, and there may be compatibility issues with different car models and brands. Additionally, the system's dependence on high-quality, continuous data streams means that any interruptions or malfunctions in the hardware could render it ineffective. Lastly, driver behaviour is highly variable and context-dependent. A system may misinterpret legitimate actions, such as glancing at a rearview mirror or checking the speedometer, as distractions. Creating algorithms that can accurately discern between safe driving practices and genuine distractions requires ongoing refinement and may still not achieve perfect accuracy.

CHAPTER 2

2. BACKGROUND THEORY

2.1 Exploration of background working of Impact of Distraction on The Design of Driver Monitoring Systems for Advanced Vehicles

Driver distraction is a significant factor contributing to road accidents and safety incidents. With the advent of advanced driver-assistance systems (ADAS) and autonomous vehicles, the ability to detect and mitigate driver distraction in real-time has become increasingly important.

Key Concepts

1. Driver Distraction: Refers to the diversion of attention away from activities critical for safe driving towards a non-driving activity. Distractions can be visual, manual, or cognitive.

2.Real-time Detection: Involves continuous monitoring and instant analysis to identify signs of distraction and provide immediate feedback or intervention.

Technological Principles Behind Detection Systems

The technologies used for detecting driver distraction rely on various principles from computer vision, sensor technology, and machine learning.

1. Computer Vision:

- **Facial Recognition**: Identifying facial landmarks and tracking movements to detect visual and cognitive distraction.
- **Eye Tracking:** Monitoring eye movements and blink patterns to determine if the driver is focused on the road**.**
- **Head Pose Estimation:** Using algorithms to analyse the orientation of the driver's head to infer attention levels.

2. Sensor Technology:

Infrared Sensors:

- Used to track eye and facial movements in low-light conditions.

Steering Wheel Sensors:

- Detecting erratic or inconsistent steering behaviour as a sign of distraction.

Seat Sensors:

- Monitoring posture changes that may indicate cognitive or manual distraction.

3. Machine Learning:

Classification Algorithms:

- Training models to classify behaviours as attentive or distracted based on data inputs from sensors and cameras.

Pattern Recognition:

- Identifying patterns in driving behaviour that correlate with distraction.

Real-time Processing:

- Implementing algorithms that can process data on-the-fly to provide immediate feedback or intervention.

<u>Methodologies for Detection and Analysis</u>

1. **Data Collection:** Gathering data from various sensors and cameras installed in the vehicle to monitor driver behaviour.
2. **Feature Extraction:** Identifying relevant features from the collected data that can indicate distraction (e.g., gaze direction, head movements, hand positions).
3. **Model Training:** Using supervised learning techniques to train models on labelled datasets where instances of distraction and non-distraction are annotated.
4. **Real-time Analysis:** Deploying trained models to analyse data in real-time and detect distraction, triggering alerts or corrective actions.

<u>Applications and Integration</u>

- **Driver Monitoring Systems (DMS):** Systems specifically designed to monitor driver attention and alertness, often integrated into modern vehicles.
- **Advanced Driver-Assistance Systems (ADAS):** Enhancing existing ADAS with distraction detection capabilities to improve overall safety.
- **Autonomous Vehicles**: Ensuring that drivers remain attentive in semi-autonomous driving scenarios where human intervention might still be necessary.

Challenges and Considerations

1. **Accuracy and Reliability:**

Ensuring that detection systems have low false-positive and false-negative rates.

2. **Privacy Concerns**: Addressing issues related to the collection and processing of personal data.

3. **User Acceptance:** Ensuring that drivers are willing to use and trust these systems.

4. **Regulatory Compliance:** Adhering to laws and regulations governing the use of monitoring technologies in vehicles.

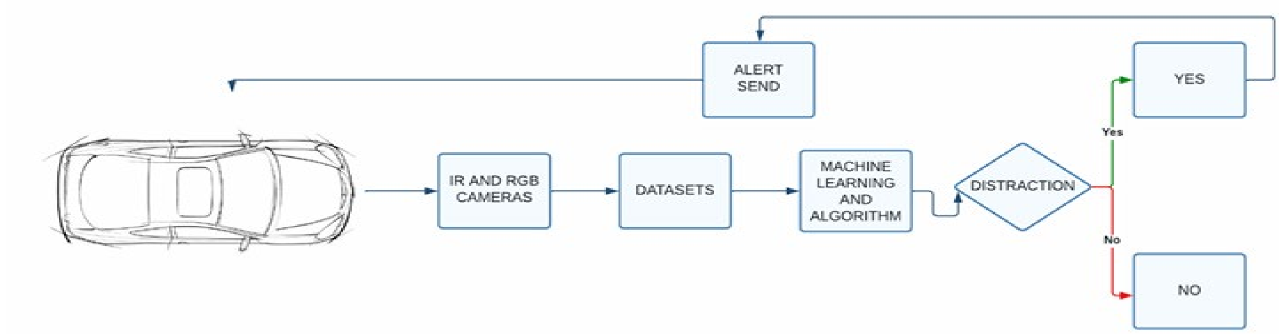


Fig 1.1 Block diagram of Impact of Distractions on The Design of a Driver Monitoring Systems for Advanced Vehicles

CHAPTER 3

3.1 LITERATURE REVIEW

Distracted driving is a major safety concern, contributing to a significant portion of traffic accidents. To address this issue, researchers have been actively developing real-time detection systems to identify driver distraction.

Li et al. (2019) [1] propose a CNN-based system in IEEE Transactions on Intelligent Transportation Systems that analyzes facial features and head pose to detect driver distraction with high accuracy.

• Vehicle Dynamics-based Systems: These systems analyze vehicle data (steering wheel movements, sudden braking) to infer driver distraction. This approach offers an alternative to visual cues, but might miss distractions not impacting vehicle control [4].

Nassif, A. B., et al. (2021). Driver Distraction Detection Using Deep Learning: State-of-the-Art and

Perspectives. IEEE Transactions on Intelligent Transportation Systems.

• This comprehensive review covers the state-of-the-art methods in driver distraction detection Using deep learning, providing a broader context for YOLOv5's application.

Wang et al. (2018) [5] in Sensors (MDPI) explore using EEG signals to detect driver fatigue, demonstrating the potential of physiological data.

<u>Machine Learning Approaches</u>:

• Traditional ML algorithms like Support Vector Machines (SVMs) were initially used with some success [6].

• Deep learning approaches using convolutional neural networks (CNNs) are now widely adopted due to their superior performance in handling complex visual data and achieving higher accuracy in identifying distractions [2, 3].

Masood et al. (2018) [2] in Pattern Recognition Letters propose a CNN-based system for driver distraction detection, showcasing the effectiveness of deep learning.

Challenges and Considerations:

• **Accuracy and Generalizability:** Developing models with high accuracy across diverse demographics, lighting conditions, and driving scenarios remains a challenge. This is an active area of research.

• **Privacy Concerns**: In-vehicle camera systems raise privacy concerns regarding data collection and storage. Balancing functionality with user privacy is crucial.

• **Alert Fatigue:** Frequent false alarms from the system can lead to driver desensitization and ignoring real warnings.

Future Directions:

• Integration of different detection methods (vision, vehicle dynamics, physiological data) is being explored for a more robust and comprehensive approach.

• Advancements in explainable AI can improve user trust and understanding of the system's alerts.

• Integrating driver distraction detection systems with Advanced Driver-Assistance Systems (ADAS) holds promise for semi-autonomous and autonomous vehicles.

"Driver Detection Using Deep Learning Based on IR Camera Images" by Wang et al. (2022):

• Wang et al. propose a deep learning-based approach for driver fatigue detection using IR camera images. They utilize convolutional neural networks (CNNs) to extract features from IR images and classify fatigue states. The proposed method achieves high accuracy and robustness in detecting driver fatigue under various conditions.

Zhou, F., & Xu, T. (2019). Driver Distraction Detection Using Camera and Infrared Sensors. Sensors, 19(18), 3910.

• The article discusses the use of camera and infrared sensors for detecting driver distraction, aligning with the suggested use of Azure RGB and IR cameras.

Liu, Z., et al. (2019). Deep Learning for Detecting Driver Distraction Based on Object Detection. IEEE Access, 7, 13907-13917.

• This paper explores deep learning techniques for detecting driver distraction, demonstrating the application of object detection frameworks like YOLO in automotive safety.

CHAPTER 4

PROBLEM STATEMENT

4.1 Problem Statement

- Driver distraction is a significant cause of road accidents, leading to injuries and fatalities. With the proliferation of in-car technologies and mobile devices, the risk of distractions has increased manifold. Traditional methods for detecting driver distraction lack real-time monitoring capabilities and may not be sufficiently accurate.
- There is a pressing need for a real-time detection system that can accurately identify instances of driver distraction and alert the driver to prevent accidents**.**

4.2 Objectives

- **Develop Reliable Detection Systems:** To create accurate and reliable systems capable of detecting various forms of driver distraction, such as texting, talking on the phone, or interacting with in-car systems.
- **User-Friendly Interface**: To design an intuitive and non-intrusive user interface that provides clear alerts to the driver.
- **Adapt to Different Driving Conditions:** To ensure the system can operate effectively under various driving conditions, including different weather, lighting, and traffic scenarios.

4.3 Methodology

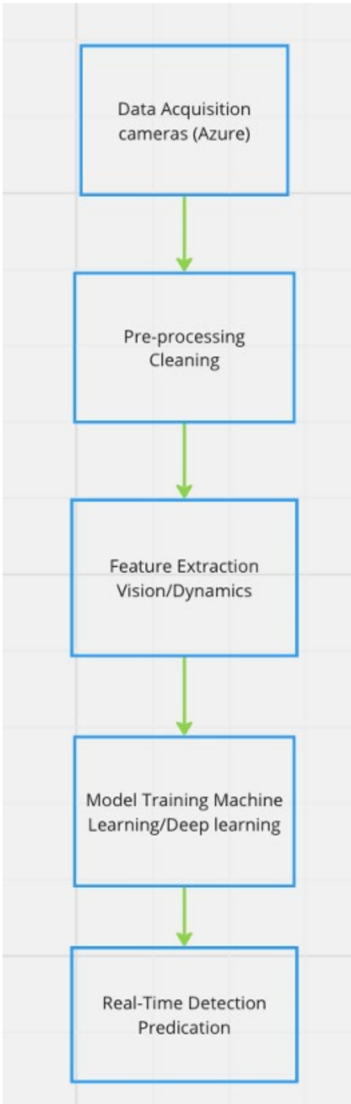


Fig 1.2 Flow chart of Impact of Distraction on The Design of a Driver Monitoring Systems for Advanced Vehicles

Data Acquisition:

• In-vehicle camera to capture driver behaviour.

• Optional: Sensors for physiological data (heart rate, EEG) or vehicle dynamics (steering wheel, braking). Preprocessing:

• Image or sensor data cleaning and normalization.

• Region of interest (ROI) selection focusing on the driver's face and head.

Feature Extraction:

• **Vision-based systems:** Facial landmarks, eye gaze direction, head pose, hand movements.
• **Vehicle dynamics-based systems:** Steering wheel angle, braking patterns, lane departure.
• **Physiological data-based systems**: Heart rate variability, EEG features.

Model Training:

• Selection of a machine learning or deep learning algorithm (e.g., CNN, SVM).
• Training the model on labeled data sets where driver behavior is categorized as "distracted" or "not distracted."

Real-Time Detection:

- The trained model analyzes new data streams from the camera or sensors in real-time.
- The model outputs a prediction indicating the likelihood of driver distraction.

CHAPTER 5

SOFTWRAE DESIGN OF PROJECT THEME

5.1 Data collection

1. Imports

```
from initializer import initialize_details, file_constructor, ImageAnnotator, add_comments_ir_rgb
import os
import time
import pykinect_azure as pykinect
import cv2
import numpy as np
import keyboard
```

The initial comment block provides metadata about the code

. **Various modules are imported:** initializer for initializing and constructing file paths, os for directory operations, time for time-related functions, pykinect_azure for interfacing with Azure Kinect devices, cv2 for OpenCV functions, numpy for numerical operations, and keyboard for capturing keyboard events.

2. Configuration and Initialization

```
rotate_flag = 1
annotations_flag = 0
number_of_subjects = 1
class_names = ['FOCUSED', 'SLEEPY', 'DISTRACTED']
time_to_capture = 10
file_extension = 'jpeg'
file_extension_annotations = 'txt'
pykinect.initialize_libraries()
device_config = pykinect.default_configuration
device_config.color_resolution = pykinect.K4A_COLOR_RESOLUTION_720P
device_config.color_format = pykinect.K4A_IMAGE_FORMAT_COLOR_YUY2
device_config.depth_mode = pykinect.K4A_DEPTH_MODE_WFOV_2X2BINNED
device_config.camera_fps = pykinect.K4A_FRAMES_PER_SECOND_30
```

```
device = pykinect.start_device(config=device_config)
```

Various flags and configuration settings are initialized:

- rotate_flag indicates if the images should be rotated by 180 degrees.
- annotations_flag determines if the annotation process should be enabled.
- number_of_subjects specifies the number of subjects in the frame.
- class names list the possible distraction classes.
- Time to capture sets the duration for capturing data.
- file_extension and file_extension_annotations specify the file formats for saved data and annotations.

• The pykinect library is initialized, and the Kinect device configuration is set, including resolution, format, depth mode, and frame rate.

3. Path Initialization

```
sensor_1 = 'azure_ir'

sensor_2 = 'azure_rgb'


path_ir = initialize_details(sensor_1)

path_rgb = path_ir.replace(sensor_1, sensor_2)

os.makedirs(path_rgb, exist_ok=True)
```

Paths for saving IR and RGB data are initialized using initialize_details. The path_rgb is derived from path_ir.

3. Annotation Process

```
disturbance = ''

if annotations_flag:
    ir_annotation_string = []
    counter = 0
    while counter < number_of_subjects:
        annotator = ImageAnnotator()
        annotator.set_class_names(class_names)

        capture_anno = device.update()
        ret, ir = capture_anno.get_ir_image()
        ir = ir.astype(np.int32)

        if not ret:
            pass
```

```
        if rotate_flag:
            ir = cv2.rotate(ir, cv2.ROTATE_180)

        adjusted_anno = cv2.convertScaleAbs(ir, alpha=alpha, beta=beta)
        annotator.annotate_frame(adjusted_anno)
        ir_annotation_string.append(annotator.annotation_string)
        print(ir_annotation_string)
        counter += 1

if annotations_flag:
    counter = 0
    rgb_annotation_string = []
    while counter < number_of_subjects:
        annotator = ImageAnnotator()
        annotator.set_class_names(class_names)

        capture_anno = device.update()
        ret, rgb = capture_anno.get_color_image()

        if not ret:
            pass

        if rotate_flag:
            rgb = cv2.rotate(rgb, cv2.ROTATE_180)

        annotator.annotate_frame(rgb)
        rgb_annotation_string.append(annotator.annotation_string)
        print(rgb_annotation_string)
        counter += 1
```

• If annotation is enabled (annotations_flag), the script collects and annotates frames from both IR and RGB cameras.

• Image Annotator is used to annotate the frames with specified class names.

4. Main Data Collection Function

```
def azure_data():
  frame_count = 0
  cv2.namedWindow('Infrared Image', cv2.WINDOW_NORMAL)
  cv2.namedWindow('RGB Image', cv2.WINDOW_NORMAL)
  start_time = time.time()
```

```
  while True:
    capture = device.update()
     ret, ir_image = capture.get_ir_image()
    ir_image = ir_image.astype(np.int32)
ret_rgb, rgb_image = capture.get_color_image()
f not ret or not ret_rgb:
      continue
    name_i = file_constructor()
    path_i = os.path.join(path_ir, name_i)
    file_name_i = f'{path_i}_{frame_count:07d}.{file_extension}'
    data_i = os.path.join(path_i, file_name_i)
    path_r = os.path.join(path_rgb, name_i)
    file_name_r = f'{path_r}_{frame_count:07d}.{file_extension}'
    data_r = os.path.join(path_r, file_name_r)
    print(data_i)
    print(data_r)
    if rotate_flag:
      rgb_image = cv2.rotate(rgb_image, cv2.ROTATE_180)
      ir_image = cv2.rotate(ir_image, cv2.ROTATE_180)
    adjusted_ir = cv2.convertScaleAbs(ir_image, alpha=alpha, beta=beta)
    cv2.imshow('IR Image', adjusted_ir)
    cv2.imshow('RGB Image', rgb_image)
    if file_extension != 'npy':
      cv2.imwrite(data_i, adjusted_ir)
      cv2.imwrite(data_r, rgb_image)
    else:
```

```
    np.save(data_i, ir_image)

    np.save(data_r, rgb_image)

  if annotations_flag:

    anno_file = f'{path_r}_{frame_count:07d}.{file_extension_annotations}'

    anno_data = os.path.join(path_r, anno_file)

    with open(anno_data, 'w') as file:

      if type(rgb_annotation_string) == list:

        for i in range(0, number_of_subjects):

          file.write(rgb_annotation_string[i] + '\n')

      else:

        file.write(rgb_annotation_string)

  if annotations_flag:

    anno_file1 = f'{path_i}_{frame_count:07d}.{file_extension_annotations}'

    anno_data1 = os.path.join(path_i, anno_file1)

    with open(anno_data1, 'w') as file:

      if type(ir_annotation_string) == list:

        for i in range(0, number_of_subjects):

          file.write(ir_annotation_string[i] + '\n')

      else:

        file.write(ir_annotation_string)

  frame_count += 1

  if cv2.waitKey(1) == ord('q'):

    break

  if time_to_capture != 0:

    if time.time() - start_time >= time_to_capture:

      fps = frame_count / (time.time() - start_time)
```

```
            print(time.time() - start_time)

            print(f'FPS: {fps}')

            comment = input('Enter Comments:')

            add_comments_ir_rgb(comment, road_condition, traffic_condition, disturbance, s_list)

            exit()

    if cv2.waitKey(1) == ord('s'):

        fps = frame_count / (time.time() - start_time)

        print(time.time() - start_time)

        print(f'FPS: {fps}')

        comment = input('Enter Comments:')

        add_comments_ir_rgb(comment, road_condition, traffic_condition, disturbance, s_list)

        exit()

azure_data()
```

• The azure data function captures and processes frames from the Azure Kinect device.

• It sets up windows to display the IR and RGB images.

• Captures frames in a loop, processes them, constructs filenames, and saves the images.

• Annotates the images if annotations_flag is set.

• Tracks the frame count and calculates the frames per second (FPS).

• The loop can be terminated by pressing 'q' or 's', or after the specified capture time (time_to_capture)

5.2 YOLO V5 ALGORITHM

1. Imports and Setup

```
import argparse
import csv
import os
import platform
import sys
from pathlib import Path

import torch

FILE = Path(__file__).resolve()
ROOT = FILE.parents[0]
if str(ROOT) not in sys.path:
    sys.path.append(str(ROOT))
ROOT = Path(os.path.relpath(ROOT, Path.cwd()))
```

2.Utility Imports

```
from ultralytics.utils.plotting import Annotator, colors, save_one_box
from models.common import DetectMultiBackend
from utils.dataloaders import IMG_FORMATS, VID_FORMATS, LoadImages, LoadScreenshots,
LoadStreams
from utils.general import (LOGGER, Profile, check_file, check_img_size, check_imshow,
check_requirements, colorstr, cv2,
                        increment_path, non_max_suppression, print_args, scale_boxes, strip_optimizer,
xyxy2xywh)
from utils.torch_utils import select_device, smart_inference_mode
```

```
@smart_inference_mode()
def run(
        weights=ROOT / 'yolov5s.pt',
        source=ROOT / 'data/images',
        data=ROOT / 'data/coco128.yaml',
        imgsz=(640, 640),
        conf_thres=0.25,
        iou_thres=0.45,
        max_det=1000,
        device='',
        view_img=False,
        save_txt=False,
        save_csv=False,
        save_conf=False,
        save_crop=False,
        nosave=True,
```

```
        classes=None,
        agnostic_nms=False,
        augment=False,
        visualize=False,
        update=False,
        project=ROOT / 'runs/detect',
        name='exp',
        exist_ok=False,
        line_thickness=3,
        hide_labels=False,
        hide_conf=False,
        half=False,
        dnn=False,
        vid_stride=1,
):
```

2. Model Loading

```
    device = select_device(device)
    model = DetectMultiBackend(weights, device=device, dnn=dnn, data=data, fp16=half)
    stride, names, pt = model.stride, model.names, model.pt
    imgsz = check_img_size(imgsz, s=stride)
```

3. Data Loading

```
    bs = 1
    if webcam:
        view_img = check_imshow(warn=True)
        dataset = LoadStreams(source, img_size=imgsz, stride=stride, auto=pt, vid_stride=vid_stride)
        bs = len(dataset)
    elif screenshot:
        dataset = LoadScreenshots(source, img_size=imgsz, stride=stride, auto=pt)
    else:
        dataset = LoadImages(source, img_size=imgsz, stride=stride, auto=pt, vid_stride=vid_stride)
    vid_path, vid_writer = [None] * bs, [None] * bs
```

4. Inference and Post-processing

```
model.warmup(imgsz=(1 if pt or model.triton else bs, 3, *imgsz))
seen, windows, dt = 0, [], (Profile(), Profile(), Profile())
for path, im, im0s, vid_cap, s in dataset:
    with dt[0]:
        im = torch.from_numpy(im).to(model.device)
        im = im.half() if model.fp16 else im.float()
        im /= 255
        if len(im.shape) == 3:
            im = im[None]

    with dt[1]:
```

```
        visualize = increment_path(save_dir / Path(path).stem, mkdir=True) if visualize else False
        pred = model(im, augment=augment, visualize=visualize)

    with dt[2]:
        pred = non_max_suppression(pred, conf_thres, iou_thres, classes, agnostic_nms, max_det=max_det)

    # CSV Writing Function
    def write_to_csv(image_name, prediction, confidence):
        data = {'Image Name': image_name, 'Prediction': prediction, 'Confidence': confidence}
        with open(csv_path, mode='a', newline='') as f:
            writer = csv.DictWriter(f, fieldnames=data.keys())
            if not csv_path.is_file():
                writer.writeheader()
            writer.writerow(data)

    for i, det in enumerate(pred):
        seen += 1
        if webcam:
            p, im0, frame = path[i], im0s[i].copy(), dataset.count
            s += f'{i}: '
        else:
            p, im0, frame = path, im0s.copy(), getattr(dataset, 'frame', 0)

        p = Path(p)
        save_path = str(save_dir / p.name)
        txt_path = str(save_dir / 'labels' / p.stem) + ('' if dataset.mode == 'image' else f'_{frame}')
        s += '%gx%g ' % im.shape[2:]
        gn = torch.tensor(im0.shape)[[1, 0, 1, 0]]
        imc = im0.copy() if save_crop else im0
        annotator = Annotator(im0, line_width=line_thickness, example=str(names))
        if len(det):
            det[:, :4] = scale_boxes(im.shape[2:], det[:, :4], im0.shape).round()

            for c in det[:, 5].unique():
                n = (det[:, 5] == c).sum()
                s += f"{n} {names[int(c)]}{'s' * (n > 1)}, "

            for *xyxy, conf, cls in reversed(det):
                c = int(cls)
                label = names[c] if hide_conf else f'{names[c]}'
                confidence = float(conf)
                confidence_str = f'{confidence:.2f}'

                if save_csv:
                    write_to_csv(p.name, label, confidence_str)

                if save_txt:
                    xywh = (xyxy2xywh(torch.tensor(xyxy).view(1, 4)) / gn).view(-1).tolist()
                    line = (cls, *xywh, conf) if save_conf else (cls, *xywh)
                    with open(f'{txt_path}.txt', 'a') as f:
                        f.write(('%g ' * len(line)).rstrip() % line + '\n')
```

```
if save_img or save_crop or view_img:
    label = None if hide_labels else (names[c] if hide_conf else f'{names[c]} {conf:.2f}')
    annotator.box_label(xyxy, label, color=colors
```

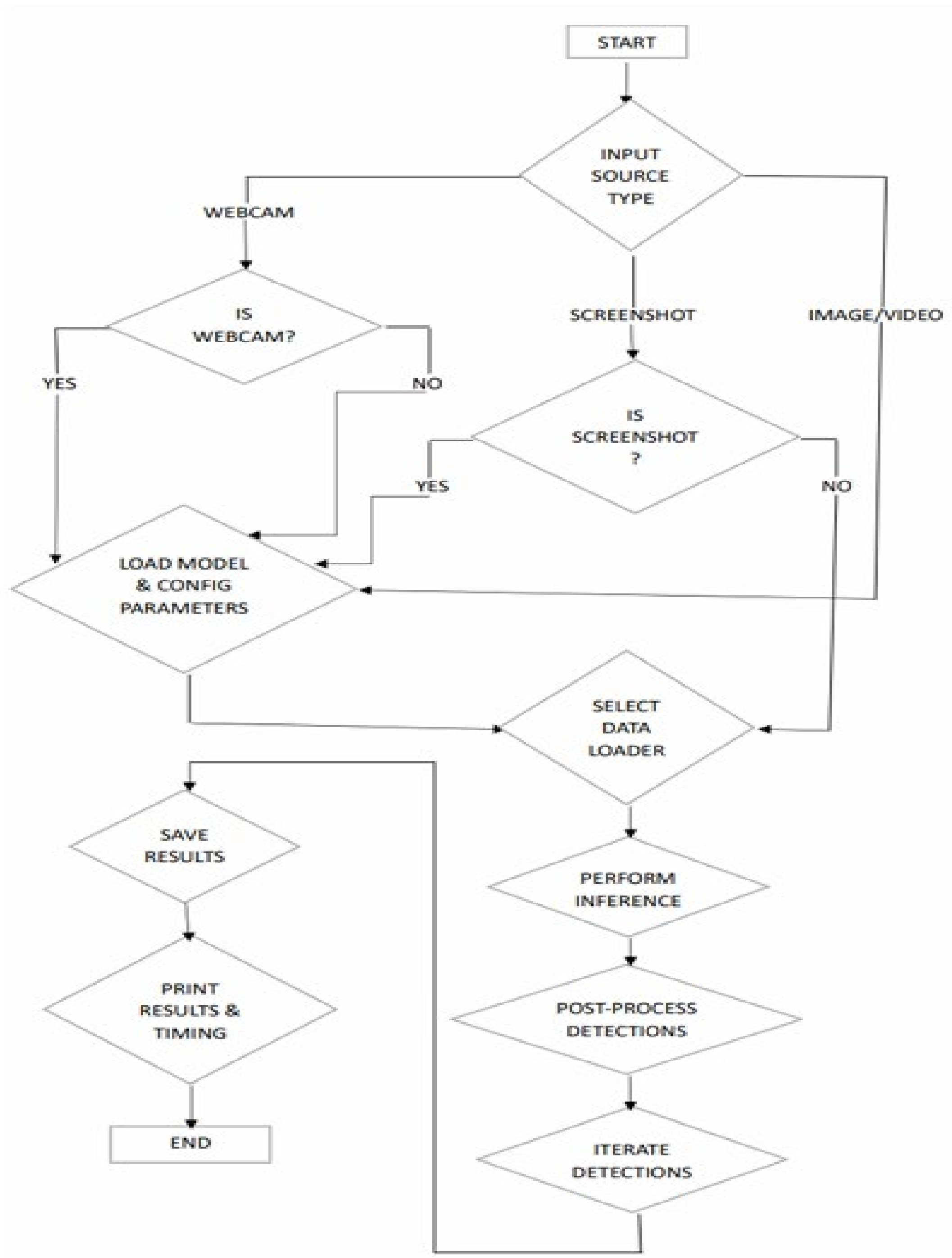


Fig 1.3 Flow Chart of YOLO V5 Algorithm

CHAPTER 6

6. RESULTS AND DISCUSSION

Confusion Matrix: -

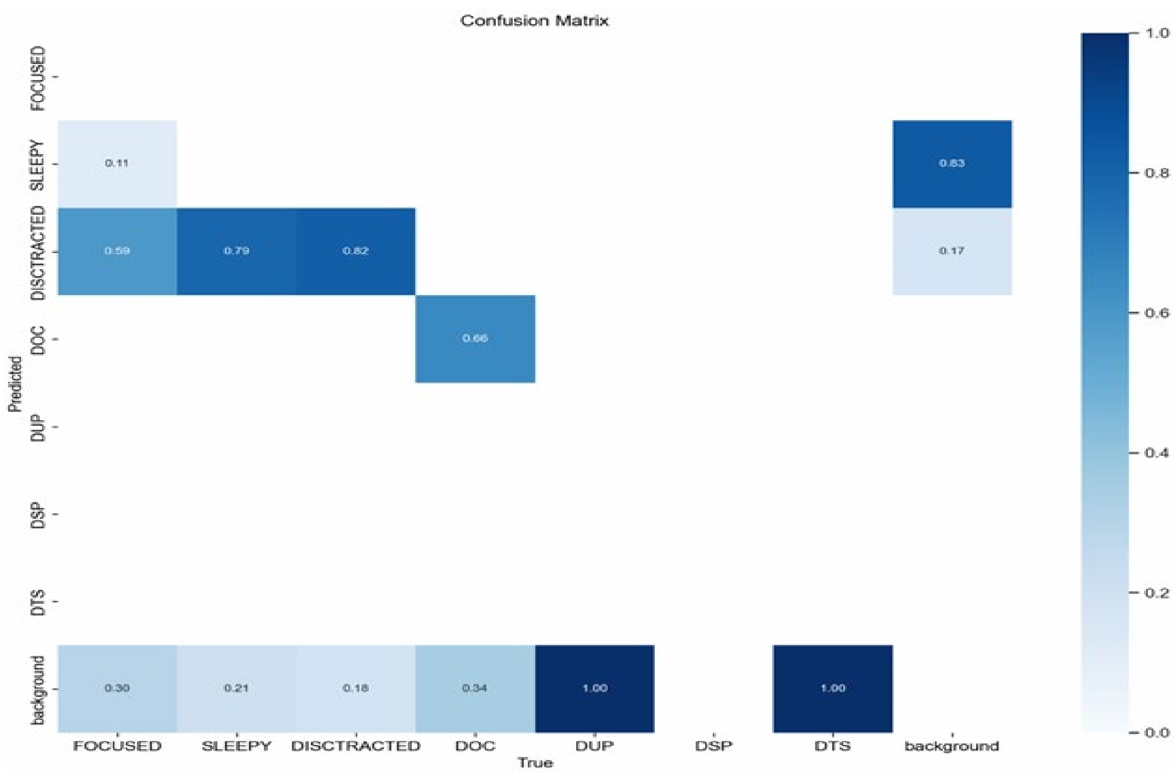


Fig 2.1 Confusion Matrix

Confusion matrix used to evaluate the performance of a deep learning model for classifying Polychlorinated Biphenyl (PCB) defects.

- Rows represent the actual classes
- Columns represent the predicted classes terms used in the confusion matrix:
- DISTRACTED: Images where the background is in focus and the PCB is not.
- DUP: “Distracted using phone.”
- DSP: “Distracted showing phone.”
- DTS: “Distracted taking selfies.”
- Background: Images where the background is in focus and the PCB is not.

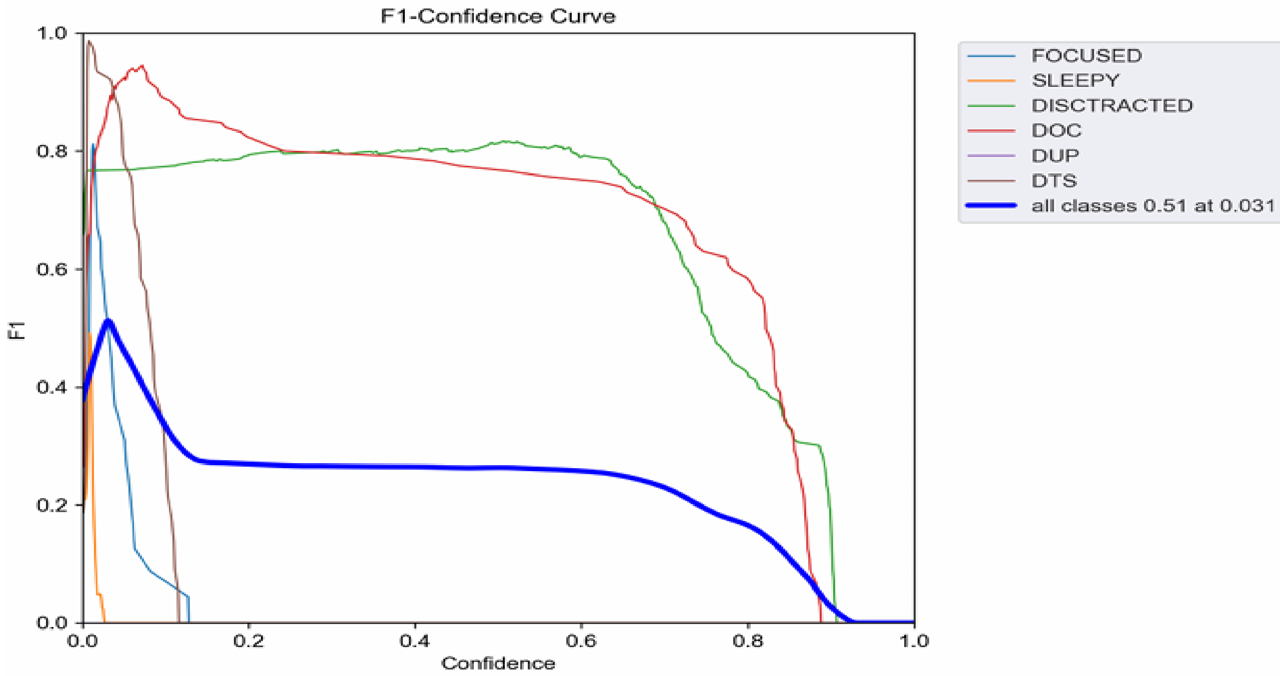


Fig 3.1 Confidence Curve

.This represents the maximum level of confidence (the model is absolutely certain about its prediction).

F1-Confidence Curve: This indicates that the curve reflects the model's confidence in achieving a good F1 score, a metric that balances precision and recall in classification tasks.

Distracted On Call Left, Distracted On Call Right, Taking Selfie and Showing Phone To Others: These are the classes the model is trying to categorize images into:

DOC, DUP, DTS, DSP: These represent classes of driver distraction that the model is designed to detect:

- DOC (distracted on call): An image where the driver is likely using a mobile phone for a call.
- DUP (distraction using phone): An image where the driver is using a mobile phone in a general sense (not necessarily for a call).
- DTS (distracted taking selfies): An image where the driver is captured in the act of taking a selfie.
- DSP (distraction showing phone): An image where the driver is holding or displaying their phone, potentially interacting with it in a distracting manner.

All classes 0.51 at 0.031: This text box signifies the model's overall performance across all classes. It achieved an average confidence level of 0.51 at a specific point (0.031) on the x-axis, which represents a certain threshold for the model's decision-making.

Confidence (y-axis): This indicates the level of certainty the model has in its classifications (ranging from 0, meaning no confidence, to 1, indicating absolute confidence). 0.0 to 1.0 on the x-axis: This represents the range of confidence values that the model can assign to an image classification.

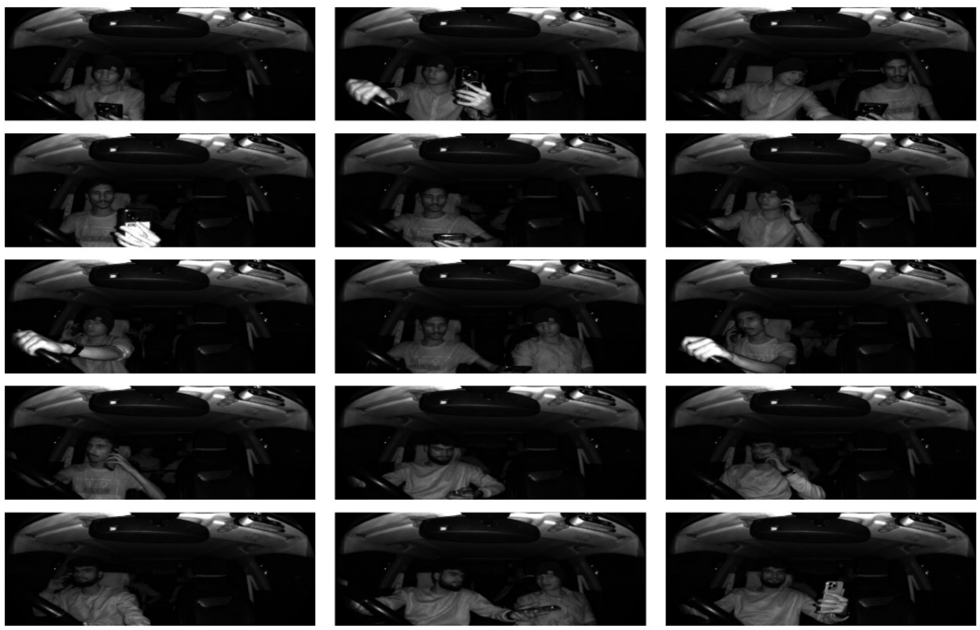
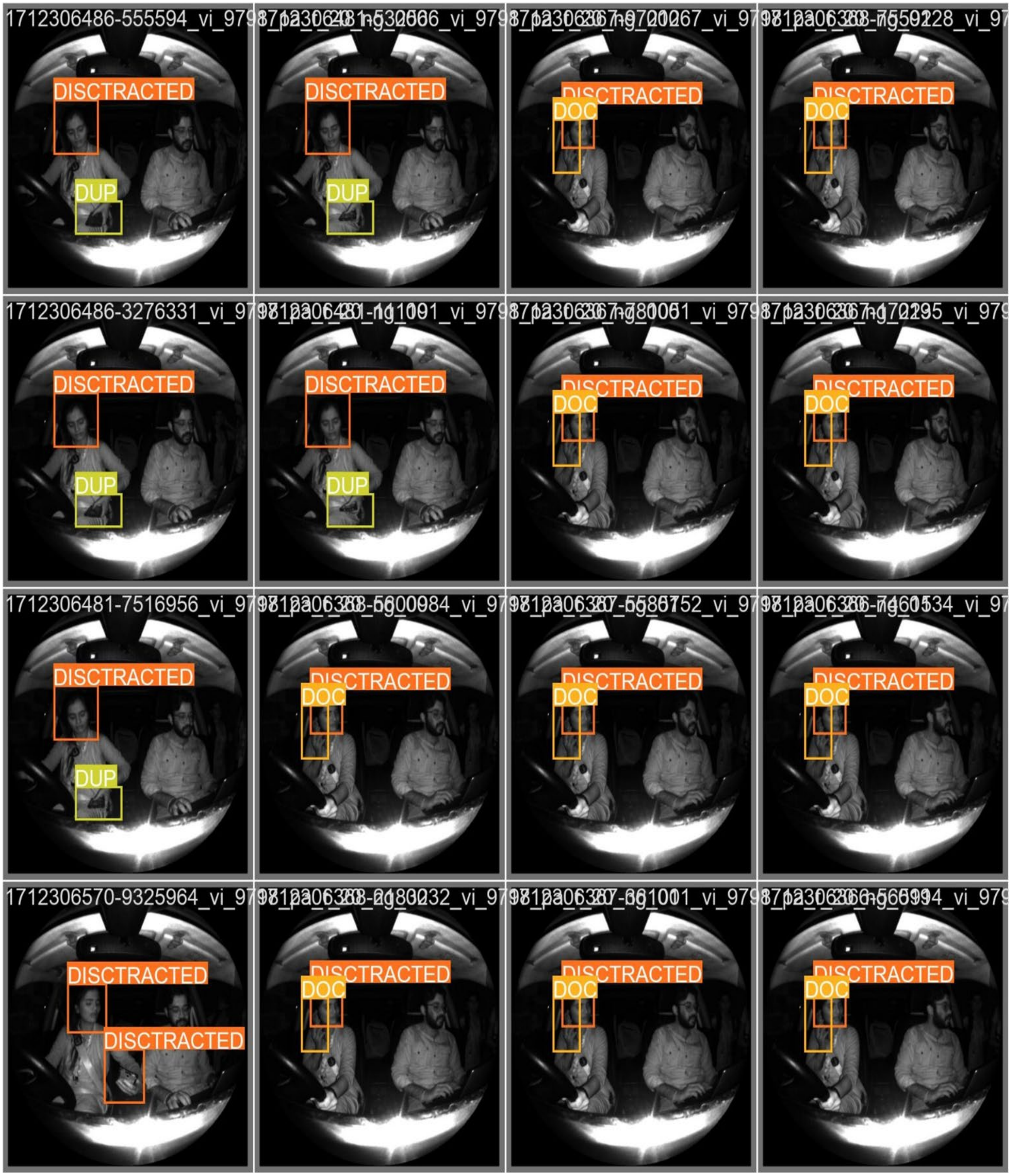


Fig 3.2 Data Collection for Different Classes


1712306486-555594_vi_97
1712306486-3276331_vi_97
1712306481-7516956_vi_97
1712306570-9325964_vi_97
DISCTRACTED
DUP
DOC

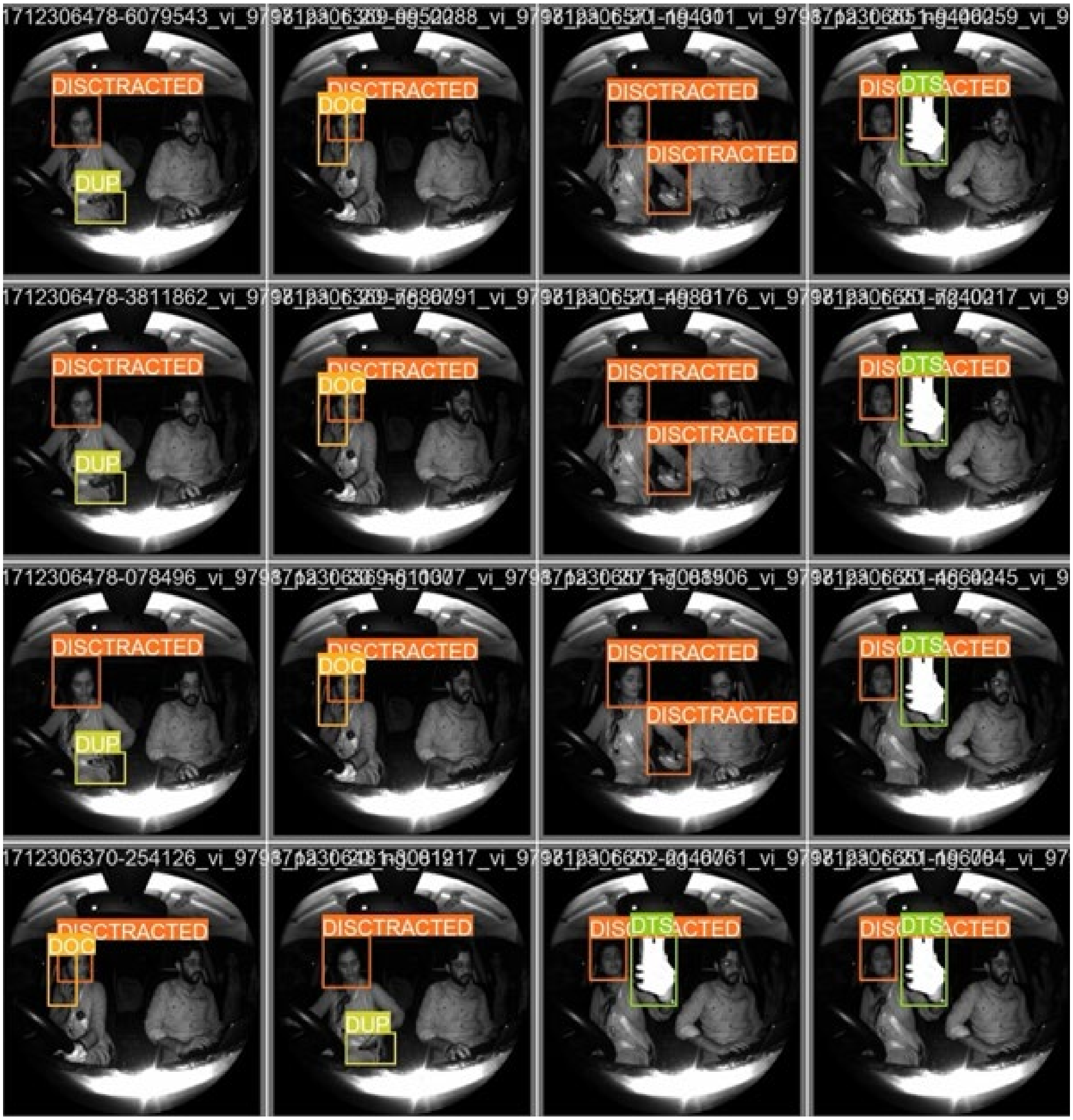


Fig 3.3 Model Predicted Output

The above image shows the predication of the classes done by the YOLO V5 model were DUP means Distracted using phone, DOC means Distracted on call, DSP means distracted showing phone, DTS Distracted taking selfies.

CHAPTER 7
CONCLUSION AND FUTURE SCOPE

The implementation of YOLOv5 for Impact of Distraction on The Design of Driver Monitoring Systems for Advanced Vehicles demonstrates significant advancements in enhancing vehicle safety. By leveraging the state-of-the-art capabilities of YOLOv5, the system can accurately detect various distractions, such as using a phone or not looking at the road, in real-time. This timely detection allows for immediate alerts to the driver, potentially preventing accidents. The high accuracy and versatility of YOLOv5, which can be deployed with multiple input sources and formats, make it suitable for integration into diverse automotive systems. The efficient processing ensures minimal latency, which is crucial for real-time applications in dynamic environments like driving.

Looking ahead, there are several areas for further development. Enhancing model performance through optimization techniques, such as model quantization and pruning, can reduce computational load and improve inference speed, particularly on embedded systems. Exploring advanced architectures or newer versions of YOLO could also yield better performance and accuracy. Integrating data from multiple cameras, such as Azure RGB and infrared (IR) cameras, along with lux sensors for light intensity measurement, could improve detection reliability under varying lighting and weather conditions. Additionally, incorporating contextual information from vehicle telemetry, such as speed and steering angle, can enhance the understanding of driver behaviour.

Robustness and generalization of the model can be improved through extensive data augmentation and transfer learning, allowing adaptation to different vehicle types and interior layouts without extensive retraining. Real world testing and validation through field trials are essential to fine-tune the model based on diverse driving conditions and ensure it meets automotive safety and regulatory standards. Furthermore, developing intuitive alert systems that effectively notify drivers without causing additional distractions, and integrating the detection system with Advanced Driver Assistance Systems (ADAS) for automated interventions, can further enhance driver safety.

By focusing on these areas, Impact Of Distraction On The Design Of Driver Monitoring Systems For Advanced Vehicles can become an integral part of modern automotive safety systems, significantly reducing the risk of accidents caused by driver inattention.

CHAPTER 8

REFERENCES:

Li et al. (2019) [1] propose a CNN-based system in IEEE Transactions on Intelligent Transportation Systems that analyses facial features and head pose to detect driver distraction with high accuracy.

• Vehicle Dynamics-based Systems: These systems analyse vehicle data (steering wheel movements, sudden braking) to infer driver distraction. This approach offers an alternative to visual cues, but might miss distractions not impacting vehicle control [4].

Nassif, A. B., et al. (2021). Driver Distraction Detection Using Deep Learning: State-of-the-Art and Perspectives. IEEE Transactions on Intelligent Transportation Systems.

• This comprehensive review covers the state-of-the-art methods in driver distraction detection using deep learning, providing a broader context for YOLOv5's application.

Wang et al. (2018) [5] in Sensors (MDPI) explore using EEG signals to detect driver fatigue, demonstrating the potential of physiological data.

Machine Learning Approaches:

• Traditional ML algorithms like Support Vector Machines (SVMs) were initially used with some success [6].

• Deep learning approaches using convolutional neural networks (CNNs) are now widely adopted due to their superior performance in handling complex visual data and achieving higher accuracy in identifying distractions [2, 3].

Masood et al. (2018) [2] in Pattern Recognition Letters propose a CNN-based system for driver distraction detection, showcasing the effectiveness of deep learning.

Challenges and Considerations:

Accuracy and Generalizability: Developing models with high accuracy across diverse demographics, lighting conditions, and driving scenarios remains a challenge. This is an active area of research.

• **Privacy Concerns**: In-vehicle camera systems raise privacy concerns regarding data collection and storage. Balancing functionality with user privacy is crucial. Real-time Detection of Driver Distraction in Automotive Applications 28

• Alert Fatigue: Frequent false alarms from the system can lead to driver desensitization and ignoring real warnings.

Future Directions:

• Integration of different detection methods (vision, vehicle dynamics, physiological data) is being explored for a more robust and comprehensive approach.
• Advancements in explainable AI can improve user trust and understanding of the system's alerts.
• Integrating driver distraction detection systems with Advanced Driver-Assistance Systems (ADAS) holds promise for semi-autonomous and autonomous vehicles.

"Driver Detection Using Deep Learning Based on IR Camera Images" by Wang et al. (2022):

• Wang et al. propose a deep learning-based approach for driver fatigue detection using IR camera images. They utilize convolutional neural networks (CNNs) to extract features from IR images and classify fatigue states. The proposed method achieves high accuracy and robustness in detecting driver fatigue under various conditions.

Zhou, F., & Xu, T. (2019). Driver Distraction Detection Using Camera and Infrared Sensors. Sensors, 19(18), 3910.

• The article discusses the use of camera and infrared sensors for detecting driver distraction, aligning with the suggested use of Azure RGB and IR cameras.

Liu, Z., et al. (2019). Deep Learning for Detecting Driver Distraction Based on Object Detection. IEEE Access, 7, 13907-13917.

• This paper explores deep learning techniques for detecting driver distraction, demonstrating the application of object detection frameworks like YOLO in automotive safety.